MODEL LAB · 2026-10-18

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	25,208	3.05%	0.6079
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-05, scored on the 25,208 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5829	1.70×	0.02952	0.13558	0.00606
KNN comps Prior is the pooled rate of the nearest comparables.	0.6079	1.59×	0.02946	0.13424	0.00402
SVR Support-vector regression on the same feature matrix.	0.6017	1.52×	0.02948	0.13476	0.00534
Random forest Forest regression; a learned similarity metric.	0.6026	1.55×	0.02947	0.13434	0.00442
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6079	1.59×	0.02946	0.13424	0.00402
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6079	1.59×	0.02946	0.13424	0.00402
Linear regression Ordinary least squares on the same features.	0.6052	1.61×	0.02948	0.13453	0.00385
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6060	1.63×	0.02949	0.13456	0.00511
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6047	1.60×	0.02947	0.13439	0.00486
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6023	1.59×	0.02948	0.13452	0.00485

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **253** knn **253** svr **253** rf **253** core **253** ensemble **253** linear **253** logistic **253** nn **253** xgb **253**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,707 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5817	1.21×	0.02380	0.11477	0.01172
KNN comps Prior is the pooled rate of the nearest comparables.	0.6281	1.97×	0.02371	0.11241	0.00915
SVR Support-vector regression on the same feature matrix.	0.6009	1.82×	0.02383	0.11551	0.01643
Random forest Forest regression; a learned similarity metric.	0.6081	1.66×	0.02376	0.11259	0.01330
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6281	1.97×	0.02371	0.11241	0.00915
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6281	1.97×	0.02371	0.11241	0.00915
Linear regression Ordinary least squares on the same features.	0.6159	1.66×	0.02392	0.11436	0.01352
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6258	1.36×	0.02392	0.11390	0.01528
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6208	1.82×	0.02373	0.11255	0.01151
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6105	1.82×	0.02384	0.11356	0.01656

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Junior Caminero 12.7%, Jo Adell 12.6%, Jonathan Aranda 12.3%, Patrick Bailey 11.9%, Ryan Vilade 11.4%, Chase DeLauter 10.8%
Flat league	6/6	Junior Caminero 11.9%, Jo Adell 11.9%, Jonathan Aranda 11.4%, Patrick Bailey 11.3%, Chase DeLauter 10.9%, Ryan Vilade 10.8%
KNN comps	6/6	Junior Caminero 12.4%, Jo Adell 12.4%, Jonathan Aranda 12.4%, Patrick Bailey 11.9%, Ryan Vilade 11.4%, Chase DeLauter 10.7%
SVR	5/6	Junior Caminero 13.9%, Jo Adell 12.8%, Jonathan Aranda 11.9%, Patrick Bailey 11.8%, Ryan Vilade 11.1%, Yandy Díaz 11.0%
Random forest	6/6	Junior Caminero 12.9%, Jo Adell 12.4%, Jonathan Aranda 12.0%, Ryan Vilade 11.6%, Patrick Bailey 11.5%, Chase DeLauter 10.7%
Core (KNN+SVR)	6/6	Junior Caminero 12.4%, Jo Adell 12.4%, Jonathan Aranda 12.4%, Patrick Bailey 11.9%, Ryan Vilade 11.4%, Chase DeLauter 10.7%
Forest blend	6/6	Junior Caminero 12.4%, Jo Adell 12.4%, Jonathan Aranda 12.4%, Patrick Bailey 11.9%, Ryan Vilade 11.4%, Chase DeLauter 10.7%
Linear regression	6/6	Junior Caminero 12.7%, Jo Adell 12.6%, Jonathan Aranda 12.2%, Patrick Bailey 11.7%, Ryan Vilade 11.1%, Chase DeLauter 10.6%
Logistic regression	6/6	Junior Caminero 13.2%, Jo Adell 12.9%, Jonathan Aranda 11.9%, Patrick Bailey 11.3%, Ryan Vilade 11.0%, Chase DeLauter 10.5%
Neural net (torch)	6/6	Junior Caminero 13.2%, Jo Adell 12.8%, Jonathan Aranda 12.0%, Patrick Bailey 11.6%, Ryan Vilade 11.3%, Chase DeLauter 10.6%
XGBoost	5/6	Junior Caminero 13.2%, Jo Adell 12.7%, Jonathan Aranda 12.2%, Patrick Bailey 11.6%, Ryan Vilade 11.4%, Yandy Díaz 10.7%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	Junior Caminero TB slate	12.7%	11.9%	12.4%	13.9%	12.9%	12.4%	12.4%	12.7%	13.2%	
2	Jo Adell CLE slate	12.6%	11.9%	12.4%	12.8%	12.4%	12.4%	12.4%	12.6%	12.9%	
3	Jonathan Aranda TB	12.3%	11.4%	12.4%	11.9%	12.0%	12.4%	12.4%	12.2%	11.9%	
4	Patrick Bailey CLE	11.9%	11.3%	11.9%	11.8%	11.5%	11.9%	11.9%	11.7%	11.3%	
5	Ryan Vilade TB	11.4%	10.8%	11.4%	11.1%	11.6%	11.4%	11.4%	11.1%	11.0%	
6	Chase DeLauter CLE	10.8%	10.9%	10.7%	10.7%	10.7%	10.7%	10.7%	10.6%	10.5%	
7	Travis Bazzana CLE	10.6%	10.5%	10.3%	10.2%	10.4%	10.3%	10.3%	10.4%	10.1%	
8	Yandy Díaz TB	10.4%	10.7%	10.4%	11.0%	10.7%	10.4%	10.4%	10.6%	10.4%	
9	Nathaniel Lowe CLE	10.1%	10.0%	10.3%	10.1%	10.3%	10.3%	10.3%	10.2%	10.1%	
10	José Ramírez CLE	9.2%	9.5%	9.3%	9.4%	9.4%	9.3%	9.3%	9.6%	9.8%	
11	Petey Halpin CLE	8.4%	9.2%	8.3%	8.7%	8.7%	8.3%	8.3%	9.3%	8.8%	
12	Brayan Rocchio CLE	6.4%	6.8%	6.3%	6.3%	6.0%	6.3%	6.3%	6.1%	6.1%	
13	Jonny DeLuca TB	6.0%	6.6%	5.8%	5.9%	5.4%	5.8%	5.8%	5.8%	6.0%	
14	Liam Hicks TB	5.0%	5.9%	4.9%	5.6%	4.8%	4.9%	4.9%	5.3%	5.2%	
15	Steven Kwan CLE	4.5%	7.5%	4.8%	3.1%	3.6%	4.8%	4.8%	3.1%	4.5%	
16	Ben Williamson TB	3.9%	6.5%	3.8%	3.9%	3.7%	3.8%	3.8%	2.8%	4.0%	
17	Taylor Walls TB	3.7%	5.1%	3.7%	3.1%	3.2%	3.7%	3.7%	3.9%	3.8%	
18	Chandler Simpson TB	2.1%	4.8%	2.7%	1.3%	1.3%	2.7%	2.7%	1.2%	2.4%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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